INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year

APPLIED DATA STRUCTURES & ALGORITHMS LAB (BCS553)

LAB 7

Assignment 7: Practice Set

Assignment aim: Practice array searching, sorting, multiplication and conditionals.

Experiment No. 7.1 — Search for 52 using a loop

1. Experiment Title: Search for 52 using a loop

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: Loops and if-statements can be combined to search, sort, multiply or filter array values.

5. Algorithm / Procedure:
Initialize index to -1; scan for 52; store its index and stop when found; print the appropriate message. Then compile and execute the Java program.

6. Program:

```
public class Task_01_Search_for_52_using_a_loop{
    public static void main(String[] args) {
        int[] a = {45, 52, 80};
        int index = -1;
        for (int i = 0; i < a.length; i++) {
            if (a[i] == 52) { index = i; break; }
        }
        if (index >= 0) System.out.println("Found at index " + index);
        else System.out.println("Not found");
    }
}
```

7. Input: Fixed input values are declared in the program: the fixed arrays specified in the Practice Set task.

8. Output:
Found at index 1

Time / Space Complexity: O(n) worst-case time; O(1) auxiliary space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- Why initialize index to -1?
- What does break do?
- What does the output mean?

Experiment No. 7.2 — Bubble sort the array {3, 1, 2}

1. Experiment Title: Bubble sort the array {3, 1, 2}

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: Loops and if-statements can be combined to search, sort, multiply or filter array values.

5. Algorithm / Procedure:
Compare adjacent values over repeated passes; swap out-of-order pairs; print the sorted array. Then compile and execute the Java program.

6. Program:

```
public class Task_02_Bubble_sort_the_array {
    public static void main(String[] args) {
        int[] a = {3, 1, 2};
        for (int pass = 0; pass < a.length - 1; pass++) {
            for (int i = 0; i < a.length - 1 - pass; i++) {
                if (a[i] > a[i + 1]) {
                    int temp = a[i]; a[i] = a[i + 1]; a[i + 1] = temp;
                }
            }
        }
        for (int value : a) System.out.println(value);
    }
}
```

7. Input: Fixed input values are declared in the program: the fixed arrays specified in the Practice Set task.

8. Output:
1
2
3

Time / Space Complexity: O(n²) time; O(1) auxiliary space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- What is an adjacent pair?
- Why does the inner loop shrink?
- What is the sorted result?

Experiment No. 7.3 — Multiply array values

1. Experiment Title: Multiply array values

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: Loops and if-statements can be combined to search, sort, multiply or filter array values.

5. Algorithm / Procedure:
Initialize product to 1; multiply it by each array value; print the final product.
Then compile and execute the Java program.

6. Program:

```
public class Task_03_Multiply_array_values {
    public static void main(String[] args) {
        int[] a = {2, 3, 4};
        int product = 1;
        for (int value : a) product *= value;
        System.out.println("Product: " + product);
    }
}
```

7. Input: Fixed input values are declared in the program: the fixed arrays specified in the Practice Set task.

8. Output:
Product: 24

Time / Space Complexity: O(n) time; O(1) auxiliary space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- Why initialize product to 1 instead of 0?
- What is multiplication's identity element?

Experiment No. 7.4 — Print even array values

1. Experiment Title: Print even array values

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: Loops and if-statements can be combined to search, sort, multiply or filter array values.

5. Algorithm / Procedure:
Traverse the array; test each value using remainder modulo 2; print values whose remainder is zero. Then compile and execute the Java program.

6. Program:

```
public class Task_04_Print_even_array_values{
    public static void main(String[] args) {
        int[] a = {1, 2, 3, 4};
        for (int value : a) {
            if (value % 2 == 0) System.out.println(value);
        }
    }
}
```

7. Input: Fixed input values are declared in the program: the fixed arrays specified in the Practice Set task.

8. Output:
2
4

Time / Space Complexity: O(n) time; O(1) auxiliary space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- What does % calculate?
- How do you test whether an integer is even?
- Does this program print odd values?